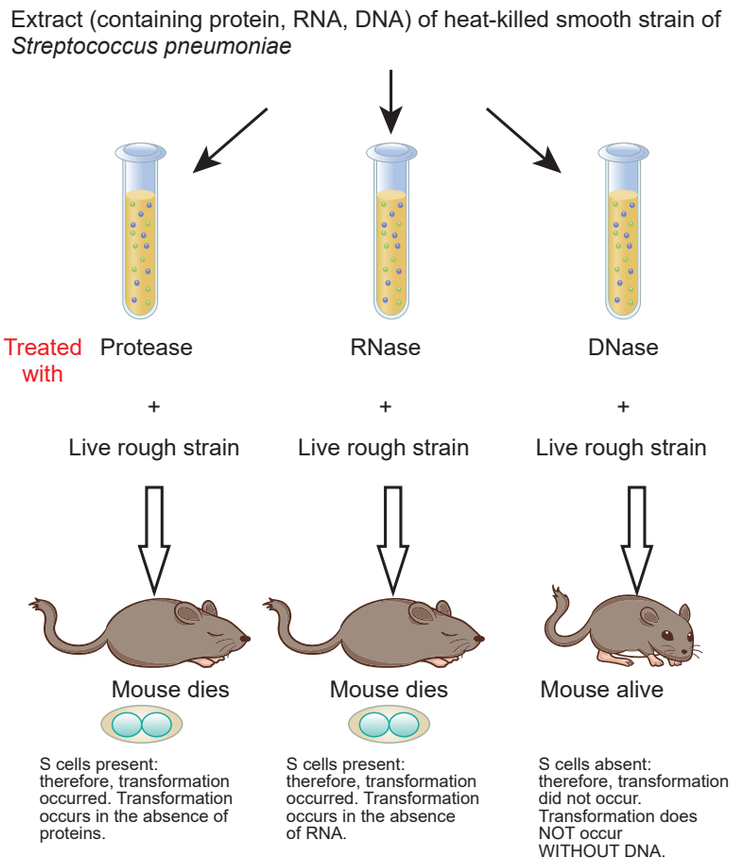


Figure 14.5: Avery, Mc Cleod and Mac Carty's experiment

Infobits

Bacterial transformation

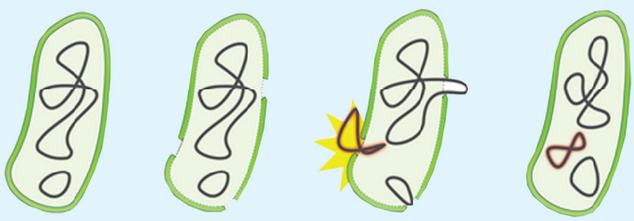
Getting a plasmid into a bacterium

Here is an E.coli. bacterium in natural state. (Notice how bacterial DNA is circular)

Extreme cold causes pores (small holes) to appear in the bacterial membrane.

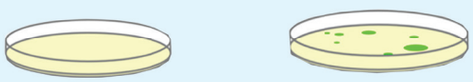
Small DNA molecules like plasmid can move through these holes.

When the bacteria are heated again, some of them end up with plasmid inside them. These are the transformed bacteria.



We can filter out the untransformed bacteria (the ones that got no plasmid) by growing all the bacteria in an antibiotic containing medium.

Untransformed bacteria are killed by the antibiotic in the medium. (They don't have the plasmid with the antibiotic resistance gene.)



The transformed bacteria grow. Now we can pick them off the plate and grow more if we want.

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